PROJECT III

SIMULATION-BASED INFERENCE USING BOOTSTRAP AND MONTE CARLO TECHNIQUE

III

SIMULATION-BASED INFERENCE
USING BOOTSTRAP AND
MONTE CARLO TECHNIQUE

Bootstrap · Monte Carlo · Statistical Simulation

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Description of Data

This study used the same data sets as Project II. The deflated total commodity price and the nominal Canada-US exchange rate will be used in this study. Because there was no recent data for the Canada-US nominal exchange rate, the study used data spanning from January 1972 to April 2017.

The objective of the study

This project aims to use simulation-based inference to revise the inferences made in project II using conventional tests. The main reason for using the simulation-based inference approach is to see if it improves on the conventional inference I performed in project II. Based on simulation-based inference, the conventional test will be used to test for cointegration, residual normality (Jarque - Bera Test), Granger - causality, residual autocorrelation, and skewness.

The theory of Simulation-Based Inference

Statistical inference consists of the use of statistics to draw conclusions about some unknown aspect of a population based on a random sample from that population. However, when making such inferences, the test statistic plays a significant role. This is because the test statistic contains information about the data that is relevant for deciding whether to reject the null hypothesis. Sometimes, we are unable to conduct such inferences due to the nature of the distribution of the test statistics (especially when the test statistic has no analytical distribution). It has been demonstrated that simulation-based techniques are more appropriate or provide better estimates for test statistics in such cases.

In general, simulation-based techniques are used when no analytical distribution for a test statistic is available, as well as when asymptotic distributions are available, but simulation is expected to improve finite–sample properties. Simulation-based techniques are used to estimate the distribution of an estimator or test statistic when the analytical distribution is unknown. Monte Carlo simulation techniques, bootstrap techniques, and resampling techniques are a few examples of simulation techniques.

The central idea behind simulation-based techniques is to create a large number of simulated samples and then use them to calculate the test statistic or estimator of interest. The calculated test statistics are used to build the empirical distribution, which

is then used to answer the question of how likely the actual test statistics are to behave under the null hypothesis.

The Kendal Density Graph in Simulation-Based – Inference

A kernel density curve may follow the shape of the distribution more closely. In short, the technique allows one to create a smooth curve given a set of random data. However, the estimation can also be used to generate points that only appear to have come from a specific sample set. This feature is particularly useful in simulation-based inference. One of the principal advantages of using kernel density graphs to represent the test statistic from a simulation is that they are better at determining the distribution shape because the number of bins used has no effect on them.

I used the kernel density graph in this project to compare the simulated test statistics to the normal density. This gives an idea of how well the simulated test statistic performs in comparison to the test statistic computed using the conventional approach to inference.

Normality Tests (Jarque-Bera test) using Simulation-Based Inference

The assumption of normality is that the underlying residuals are normally distributed, or approximately so. The normality of the residuals is an assumption made when running a linear model. So, if your residuals are normal, your assumption is correct, and your model inference (confidence intervals, model predictions) should be correct as well. If your residuals are not normal, there could be a problem with the model's fit, stability, or reliability. The Jarque - Beta test was used in the previous project to determine the residuals' normality. The test results indicated that the residuals are not normally distributed at a 5% significance level, indicating possible misspecification of the VAR model.

The conventional test will be employed here to test for normality in residuals. The test's null hypothesis states that residuals are normally distributed, while the alternate hypothesis states that residuals are not normally distributed.

The conventional Jarque-Bera test for normality shows that the null hypothesis is rejected at a 5% level of significance (see table 1.1). As a result of this result, the disturbance term is not normally distributed. Since the null hypothesis has been rejected, it is expected that bootstrap and Monte Carlo simulation-based techniques will improve the test statistic's ability to reject the null hypothesis when it is actually false.

The Jarque – Bera test for normality test is pivotal.

$$JB = \frac{n}{6}\left(S^2 + \frac{1}{4}(K-3)^2\right)$$

It does not depend on any parameters of the model. The analytical distribution in a finite sample is unknown, but the asymptotic distribution is known which is Chi-square. In this situation, the simulation-based techniques are more appropriate or provide better estimates for test statistics. The simulation results confirm that the null of normality should be rejected (Table 1.2). The conventional and simulation tests revealed that the residuals are not normally distributed, implying that the vector autoregressive model was misspecified. This means that the model does not account for all of the trends in the dataset. Both the normal distribution line and the simulated normal distribution are depicted on the kernel density graph (Table 1.3).

Conventional Approach

Selection of lag order based on AIC (Table 1.0)

```
. varsoc d.logDTCP d.logEX, exo(mon)

   Selection-order criteria
   Sample:  1972m6 - 2017m4                     Number of obs      =       539
```

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	2581.11				2.4e-07	-9.56254	-9.55009	-9.53071
1	2619.38	76.555*	4	0.000	2.1e-07*	-9.68973*	-9.66483*	-9.62606*
2	2623.06	7.3529	4	0.118	2.1e-07	-9.68853	-9.65118	-9.59303
3	2624.68	3.2504	4	0.517	2.1e-07	-9.67972	-9.62991	-9.55238
4	2629.08	8.7959	4	0.066	2.1e-07	-9.6812	-9.61894	-9.52202

```
   Endogenous:  D.logDTCP D.logEX
    Exogenous:  mon  _cons
```

Vector autoregression result using lag order of 1

```
. var d.logDTCP d.logEX, lags(1/1)

Vector autoregression

Sample:  1972m3 - 2017m4                        Number of obs     =        542
Log likelihood =   2636.389                     AIC               =  -9.706231
FPE            =   2.09e-07                     HQIC              =  -9.687638
Det(Sigma_ml)  =   2.04e-07                     SBIC              =  -9.658682

Equation           Parms      RMSE     R-sq      chi2     P>chi2
----------------------------------------------------------------
D_logDTCP              3   .035836   0.0815   48.09719   0.0000
D_logEX                3   .013904   0.0889   52.91689   0.0000
----------------------------------------------------------------

------------------------------------------------------------------------------
             |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
D_logDTCP    |
     logDTCP |
         LD. |   .2552216   .0460356     5.54   0.000     .1649935    .3454497
             |
       logEX |
         LD. |  -.1489792   .1180214    -1.26   0.207    -.3802969    .0823385
             |
       _cons |  -.0004033   .0015361    -0.26   0.793    -.0034139    .0026073
-------------+----------------------------------------------------------------
D_logEX      |
     logDTCP |
         LD. |  -.0380915    .017861    -2.13   0.033    -.0730984   -.0030845
             |
       logEX |
         LD. |   .2416676   .0457903     5.28   0.000     .1519203     .331415
             |
       _cons |   .0003813    .000596     0.64   0.522    -.0007868    .0015493
------------------------------------------------------------------------------
```

Results from the conventional test (Table 1.1)

```
. varnorm, jbera

   Jarque-Bera test
```

Equation	chi2	df	Prob > chi2
D_logDTCP	284.507	2	0.00000
D_logEX	166.073	2	0.00000
ALL	450.580	4	0.00000

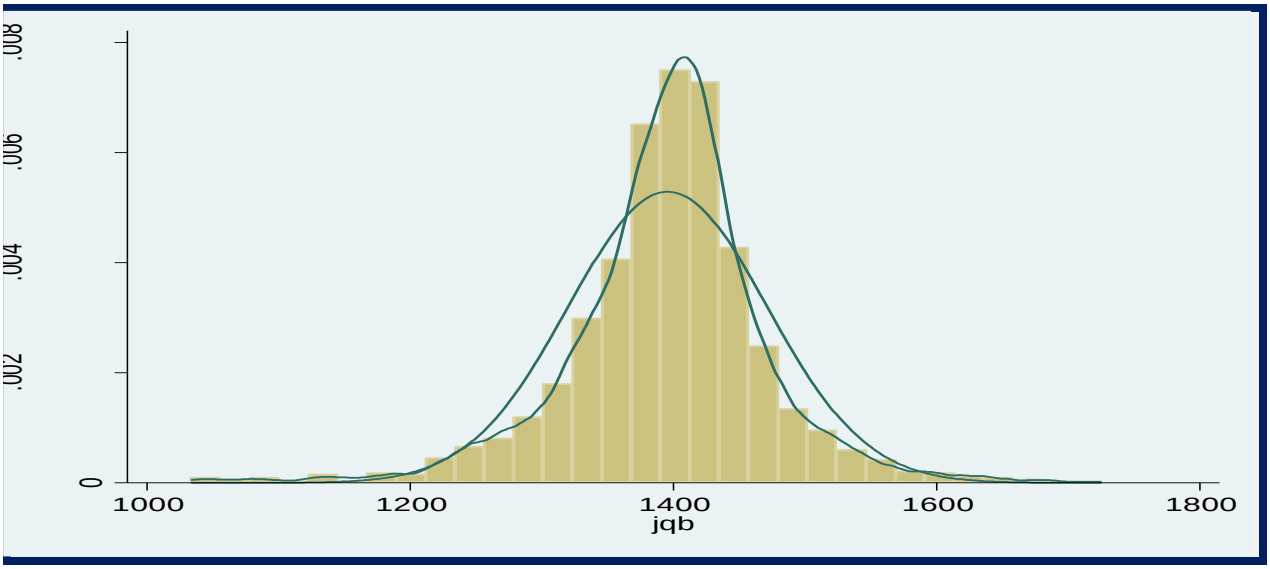
Simulation Results (Table 1.2)

```
. summarize

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
         jqb |      1,500    1395.695    75.38054   1033.403    1725.32
         rej |      1,500           1           0          1          1

. histogram jqb,normal kdensity
(bin=31, start=1033.403, width=22.319903)
```

The kernel density graph for the simulation results (Table 1.3)



Lagrange Multiplier test for residual autocorrelation using Simulation-Based Inference

The LM test for residual autocorrelation determines whether autocorrelation exists in the residuals for a given lag order. In general, the residuals from a VAR model should be white noise with no autocorrelation. If there is autocorrelation among the residuals, it indicates that some information was not accounted for by the model, such as insufficient lags. LM test for residual autocorrelation demonstrates that the VAR in the levels model is correctly specified. In the previous project (project II), I investigated whether there is autocorrelation in the residuals for a given lag order. The results showed that no autocorrelation exists in the residuals for any lag order. This shows that the VAR in the levels model was properly specified. For the purposes of this study, I will employ the conventional method of testing for autocorrelation in residuals for any lag orders estimated after the vector autoregressive model. The null hypothesis states that there is no autocorrelation in the residuals for any lag order, whereas the alternative hypothesis states that there is autocorrelation in the residuals for any lag orders estimated after the vector autoregressive model. The LM test for autoregression test is pivotal. It does not depend on any parameters of the model. The analytical distribution in a finite sample is unknown, but the asymptotic distribution is known which is Chi-square. The conventional test revealed no autocorrelation in the residuals for any lag order. Because the significance probability is greater than the significance level. (See Table 2.1). However, since the LM test statistic is asymptotically pivotal and does not depend on any parameter, we anticipate that parametric bootstrap techniques will improve the test for autocorrelation in residuals. After the simulation test, the outcome of the test revealed that there is no autocorrelation in the residuals for any lag order. Because the significance probability is greater than the significance level. (See Table 2.2). In this case, the simulation-based technique produced results, which is consistent which the conventional approach.

Conventional Approach (Table 2.0)

```
. var d.logDTCP d.logEX, lags(1/1) exog(mon)

Vector autoregression

Sample:  1972m3 - 2017m4                        Number of obs     =        542
Log likelihood =   2636.677                     AIC               =  -9.699917
FPE            =   2.10e-07                     HQIC              =  -9.675126
Det(Sigma_ml)  =   2.04e-07                     SBIC              =  -9.636518

Equation           Parms      RMSE     R-sq      chi2     P>chi2
----------------------------------------------------------------
D_logDTCP              4     .035863   0.0818    48.2969   0.0000
D_logEX                4     .013915   0.0892   53.08976   0.0000
----------------------------------------------------------------

------------------------------------------------------------------------------
             |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
D_logDTCP    |
     logDTCP |
         LD. |   .2544378   .0460642     5.52   0.000     .1641537    .3447219
             |
       logEX |
         LD. |  -.1507089   .1180705    -1.28   0.202    -.3821229     .080705
             |
         mon |  -4.21e-06   9.82e-06    -0.43   0.668    -.0000234     .000015
       _cons |   .0013485    .004369     0.31   0.758    -.0072146    .0099117
-------------+----------------------------------------------------------------
D_logEX      |
     logDTCP |
         LD. |  -.0383732   .0178725    -2.15   0.032    -.0734028   -.0033437
             |
       logEX |
         LD. |   .2410458   .0458105     5.26   0.000     .1512589    .3308326
             |
         mon |  -1.51e-06   3.81e-06    -0.40   0.691    -8.98e-06    5.95e-06
       _cons |   .0010111   .0016952     0.60   0.551    -.0023114    .0043335
------------------------------------------------------------------------------
```

LM test results (Table 2.1)

```
. varlmar, mlag (3)

    Lagrange-multiplier test
```

lag	chi2	df	Prob > chi2
1	7.1779	4	0.12678
2	8.5441	4	0.07356
3	4.0325	4	0.40163

```
    H0: no autocorrelation at lag order
```

Simulation results (Table 2.2)

```
. summarize

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
          LM |        500    67.29871     14.2565   33.08839   107.2492
         rej |        500           1           0          1          1

. histogram LM,kdensity
(bin=22, start=33.088394, width=3.3709474)
```

Cointegration test using simulation-based inference

The Augmented Dickey-Fuller test for a unit root in the residual was used to perform the conventional test for cointegration. The Augmented Dickey-Fuller test was used because there is the possibility of autocorrelation in the residuals. The null hypothesis of the test states that there is a unit root in the residuals, whereas the alternative hypothesis states that there is no unit root in the residuals. I proceed with implementing the Augmented Dickey-Fuller (ADF) Test. The test statistic in ADF showed a value of – 3.643 (Table 3.2) and the asymptotic critical value of 2.89 at a 5% level of significance. In reference to Table 3.2, the output shows that the test statistic value - 3.643 of the test statistic is smaller than the 5% asymptotic critical value of 2.89. Therefore, at a 5% level of significance, there is enough evidence to fail to reject the null hypothesis, which means we can confidently conclude that the Canada-US Nominal Spot Exchange rate and deflated commodity prices has a unit root in the residuals and hence, not cointegrated.

The simulation-based inference techniques such as bootstrap and Monte Carlo simulation were used to improve test inference. The main assumption of these techniques is that there is no analytical distribution available or that it cannot be derived. Bootstrap and Monte Carlo are expected to improve the test statistic's finite sample properties can be enhanced its power. The bootstrap and Monte Carlo simulation-based techniques are expected to improve the test statistic's ability to reject the null hypothesis when it is actually false.

Also, if the test is asymptotically pivotal, then the parametric bootstrap will be best, and it can improve inference in finite samples. Residuals will be estimated based on simulated data.

After simulation, the null hypothesis that there is a unit root in the residual was rejected, this is because it was revealed that the test statistic for the simulated distribution is greater than the test statistic for the asymptotic distribution and the significant probability from the simulated distribution is statistically. The results of the simulation-based inference show that commodity prices and the Canada-US exchange rate are cointegrated. Because the null is false, we can conclude that the simulated-based inference improved the test statistic's ability to reject the null of a unit root in the residuals. Kernel density estimation is a non-parametric method for estimating the probability density function of a random variable from a finite data sample. The kernel density, however, illustrates that the distribution is normally distributed (Table 3.4).

Conventional Approach (Table 3.0)

```
. reg logDTCP logEX mon

      Source |       SS           df       MS      Number of obs   =       544
-------------+----------------------------------   F(2, 541)       =    563.84
       Model |  26.7820389         2  13.3910194   Prob > F        =    0.0000
    Residual |  12.8485174       541   .02374957   R-squared       =    0.6758
-------------+----------------------------------   Adj R-squared   =    0.6746
       Total |  39.6305562       543   .07298445   Root MSE        =    .15411

------------------------------------------------------------------------------
     logDTCP |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       logEX |  -1.383363   .0493556   -28.03   0.000    -1.480315   -1.286411
         mon |  -.0005653   .0000428   -13.22   0.000    -.0006493   -.0004814
       _cons |   1.338811   .0197106    67.92   0.000     1.300092    1.377529
------------------------------------------------------------------------------
```

Selection – order criteria based on AIC (Table 3.1)

```
Selection-order criteria
Sample:  1972m5 - 2017m4                     Number of obs      =       540
```

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	254.701				.02288	-.939635	-.936527	-.931688
1	1070.23	1631	1	0.000	.00112	-3.95639	-3.95018	-3.9405
2	1082.69	24.932*	1	0.000	.001074*	-3.99886*	-3.98953*	-3.97502*
3	1082.72	.04753	1	0.827	.001077	-3.99524	-3.98281	-3.96345
4	1082.72	.00282	1	0.958	.001081	-3.99154	-3.976	-3.95181

```
Endogenous:  e
 Exogenous:  _cons
```

Conventional test result (Table 3.2)

```
. dfuller e, regress lag (2)

Augmented Dickey-Fuller test for unit root         Number of obs   =       541

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.643            -3.430            -2.860            -2.570
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0050

------------------------------------------------------------------------------
         D.e |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
           e |
         L1. |  -.0339225   .0093116    -3.64   0.000    -.0522141    -.015631
         LD. |   .2097781   .0428121     4.90   0.000     .1256785    .2938778
        L2D. |    .009143   .0430267     0.21   0.832    -.0753783    .0936643
             |
       _cons |   .0005649   .0014055     0.40   0.688     -.002196    .0033258
------------------------------------------------------------------------------
```

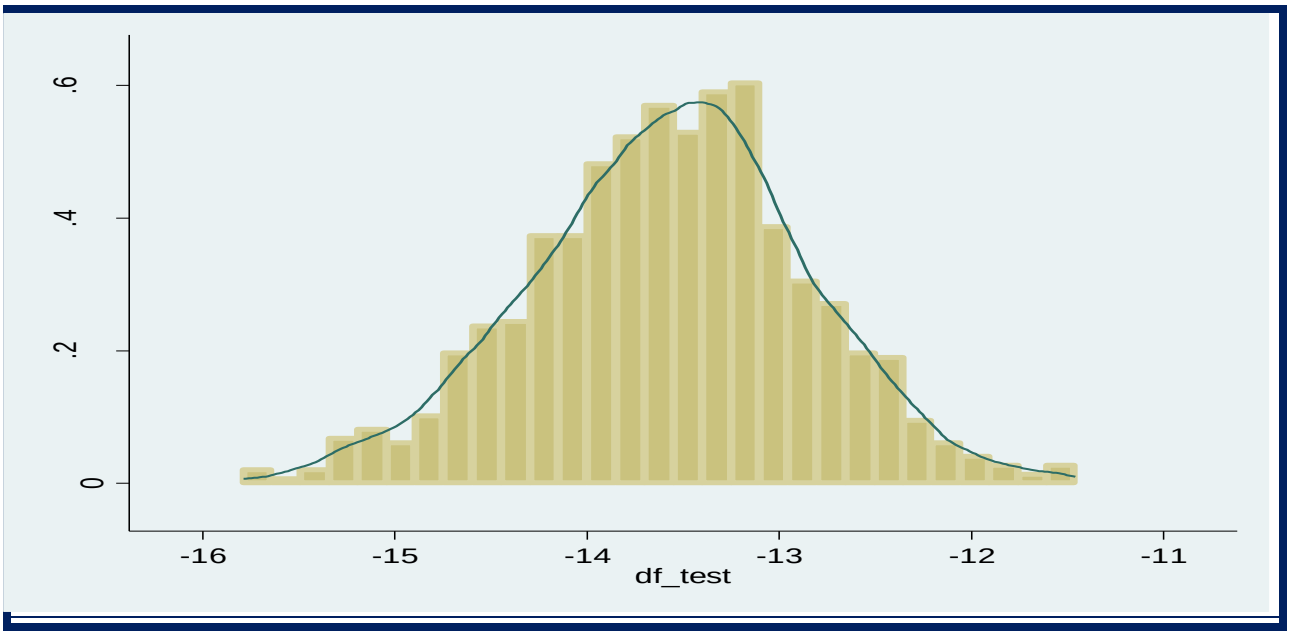
Simulation results (Table 3.3)

```
. summarize

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
     df_test |        987   -13.57424    .7176953  -15.78872  -11.45785
         rej |        987           0           0          0          0

. histogram df_test,kdensity
(bin=29, start=-15.788724, width=.14934063)
```

The kernel density graph for the simulation results (Table 3.4)



Granger Causality Test using simulation-based Inference

Granger causality is a causality concept derived from the idea that causes may not always occur aftereffects and that if one variable is the cause of another, knowing the status of the cause at an earlier point in time can help predict the effect at a later point in time (Granger, 1969; Lütkepohl 2005, p.41). The granger causality test was used in the previous project (Project II) to determine whether the Canada-US Spot Exchange Rate does not cause or has no explaining power in predicting Deflated Commodity Prices. At the 5% significance level, we fail to reject the null hypothesis, implying that the exchange rate does not granger-cause commodity prices. However, a granger causality test was performed on the normal variables again, and the results were compared to the variables in the first difference, revealing that the exchange rate granger causes commodity price. Bootstrap and Monte Carlo simulation techniques will be applied to improve inference. I then used the conventional approach to test whether Canada-US exchange rates granger-cause commodity prices, and then added one extra lag number to overfit the model for generalization and inference. The null hypothesis of the test is that the Canada-US exchange rate does not granger cause commodity prices, whereas the alternative indicates that exchange rates granger cause commodity prices. The conventional test results for the granger causality test (Table 4.1) show that the F-statistic value (3.18) is greater than the F-critical value (2.9957) at a 5% significance level, and the significant probability of 0.0422 is also statistically significant at a 5% significance level, providing enough evidence to reject the null and conclude that Canada-US exchange rates do granger-cause commodity prices. Because the null is false, we can conclude that the simulated-based inference improved the test statistic's ability to reject the null hypothesis that Canada-US exchange rates do not granger cause commodity prices. The test statistics (F - Statistic) is pivotal. This implies that the test statistic does not depend on the nuisance parameter. This is due to the fact that the F-statistic used in this analysis does not depend on other parameters and the probability of type I error will be exactly significant level.

After several simulations, the simulated test statistic and its p-value (0.037333) (Table 4.2) confirm the conventional test result. At a 5% significance level, the null hypothesis is rejected. Both tests indicate that the granger exchange rate causes deflated commodity prices. The kernel density, on the other hand, shows that the distribution is positively skewed (Table 4.3).

Conventional Approach

Selection of lag order (Table 4.0)

```
. varsoc logDTCP logEX, exo(mon)

   Selection-order criteria
   Sample:  1972m5 - 2017m4                     Number of obs      =       540
```

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	575.247				.000413	-2.11573	-2.1033	-2.08394
1	2593.14	4035.8	4	0.000	2.4e-07	-9.57459	-9.54973	-9.51101
2	2633.87	81.459*	4	0.000	2.1e-07*	-9.71063*	-9.67333*	-9.61526*
3	2637.5	7.2585	4	0.123	2.1e-07	-9.70925	-9.65952	-9.5821
4	2638.89	2.7891	4	0.594	2.1e-07	-9.6996	-9.63744	-9.54066

```
   Endogenous:  logDTCP logEX
    Exogenous:  mon  _cons
```

```
. regress logDTCP mon L(1/2).logDTCP L(1/2).logEX

      Source |       SS           df       MS      Number of obs   =       542
-------------+----------------------------------   F(5, 536)       =   6155.19
       Model |  38.9479113         5  7.78958226   Prob > F        =    0.0000
    Residual |  .67832419        536  .00126553   R-squared       =    0.9829
-------------+----------------------------------   Adj R-squared   =    0.9827
       Total |  39.6262355       541  .073246276   Root MSE        =    .03557

------------------------------------------------------------------------------
     logDTCP |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         mon |  -.0000235   .0000114    -2.06   0.040     -.000046   -1.05e-06
             |
     logDTCP |
         L1. |   1.235257   .0462774    26.69   0.000      1.14435    1.326165
         L2. |   -.267745   .0460439    -5.81   0.000    -.3581936   -.1772964
             |
       logEX |
         L1. |  -.1722598   .1180758    -1.46   0.145    -.4042078    .0596882
         L2. |   .1329381   .1177631     1.13   0.259    -.0983956    .3642719
             |
       _cons |   .0442082   .0142007     3.11   0.002     .0163123     .072104
------------------------------------------------------------------------------
```

The conventional test statistic (Table 4.1)

```
. test(L.logEX) (L2.logEX)

 ( 1)  L.logEX = 0
 ( 2)  L2.logEX = 0

       F(  2,   536) =    3.18
            Prob > F =    0.0422
```

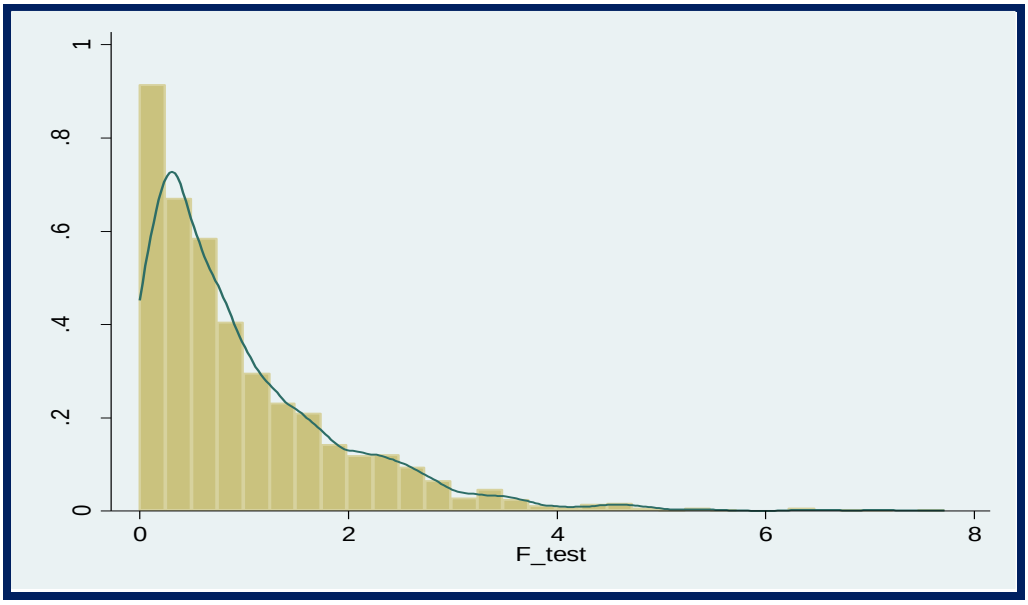
Simulation from the test statistic (Table 4.2)

```
. summarize

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
      F_test |      1,500    .9865299    .9935846   7.65e-06   7.716124
         rej |      1,500    .0373333    .1896405          0          1

. histogram F_test, kdensity
(bin=31, start=7.649e-06, width=.24890697)
```

The kernel density graph for the simulation results (Table 4.3).



Simulation-based inference for granger causality in first difference (Table 4.4)

```
. var d.logDTCP d.logEX, lags(1/1)

Vector autoregression

Sample:  1972m3 - 2017m4                        Number of obs     =        542
Log likelihood =   2636.389                     AIC               =  -9.706231
FPE            =   2.09e-07                     HQIC              =  -9.687638
Det(Sigma_ml)  =   2.04e-07                     SBIC              =  -9.658682

Equation           Parms      RMSE     R-sq      chi2     P>chi2
----------------------------------------------------------------
D_logDTCP              3     .035836   0.0815   48.09719   0.0000
D_logEX                3     .013904   0.0889   52.91689   0.0000
----------------------------------------------------------------

------------------------------------------------------------------------------
             |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
D_logDTCP    |
     logDTCP |
         LD. |   .2552216   .0460356     5.54   0.000     .1649935    .3454497
             |
       logEX |
         LD. |  -.1489792   .1180214    -1.26   0.207    -.3802969    .0823385
             |
       _cons |  -.0004033   .0015361    -0.26   0.793    -.0034139    .0026073
-------------+----------------------------------------------------------------
D_logEX      |
     logDTCP |
         LD. |  -.0380915    .017861    -2.13   0.033    -.0730984   -.0030845
             |
       logEX |
         LD. |   .2416676   .0457903     5.28   0.000     .1519203     .331415
             |
       _cons |   .0003813    .000596     0.64   0.522    -.0007868    .0015493
------------------------------------------------------------------------------
```

```
. vargranger

  Granger causality Wald tests
```

Equation	Excluded	chi2	df	Prob > chi2
D_logDTCP	D.logEX	1.5934	1	0.207
D_logDTCP	ALL	1.5934	1	0.207
D_logEX	D.logDTCP	4.5482	1	0.033
D_logEX	ALL	4.5482	1	0.033

Simulated results from the test statistic – the first difference (Table 4.5)

```
. summarize

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
      F_test |      1,500    .9865299    .9935846   7.65e-06   7.716124
         rej |      1,500    .0373333    .1896405          0          1
```

Skewness and Kurtosis test using Simulation-Based Inference

Skewness allows you to determine how much a distribution's overall shape deviates from the shape of the normal distribution. Kurtosis is a measure of how different the tails of a distribution are from the tails of a normal distribution. Kurtosis focuses on the tail shape, whereas skewness focuses on the overall shape. If skewness is equal to zero, the data is perfectly symmetrical. Real-world data is rarely if ever perfectly symmetric, instead, we get data that is approximately symmetric. The null hypothesis of the test is that the residual term has a skewness of zero. The conventional test was performed, and the results indicate that the null should be rejected at a 5% level of significance (see Table 5.2). As a result, the residual term appears to be skewed. The simulated test statistic, on the other hand, failed to reject the null of zero skewness. This means that the simulated test result indicates that the normal distribution is skewed. I used the kernel density graph, which shows that the distribution is not skewed and deviates significantly from the standard normal distribution. As a result, the conclusions from the two tests differ depending on the power of each test's test statistic.

Conventional Approach

Vector autoregression (Table 5.1)

```
Vector autoregression

Sample:  1972m4 - 2017m4                        Number of obs     =        541
Log likelihood =    2634.24                     AIC               =  -9.701441
FPE            =   2.10e-07                     HQIC              =  -9.670406
Det(Sigma_ml)  =   2.02e-07                     SBIC              =   -9.62208

Equation           Parms      RMSE     R-sq      chi2     P>chi2
----------------------------------------------------------------
D_logDTCP              5     .035847   0.0860   50.88424   0.0000
D_logEX                5     .013913   0.0925   55.11625   0.0000
----------------------------------------------------------------

------------------------------------------------------------------------------
             |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
D_logDTCP    |
     logDTCP |
         LD. |   .2464391   .0471523     5.23   0.000     .1540222     .338856
        L2D. |   .0221817   .0472906     0.47   0.639    -.0705061    .1148696
             |
       logEX |
         LD. |  -.1049544   .1212342    -0.87   0.387    -.3425692    .1326603
        L2D. |  -.1486392   .1210338    -1.23   0.219    -.3858612    .0885828
             |
       _cons |  -.0003591   .0015355    -0.23   0.815    -.0033687    .0026505
-------------+----------------------------------------------------------------
D_logEX      |
     logDTCP |
         LD. |  -.0362533   .0183006    -1.98   0.048    -.0721219   -.0003847
        L2D. |    -.01433   .0183543    -0.78   0.435    -.0503038    .0216438
             |
       logEX |
         LD. |   .2553083   .0470531     5.43   0.000     .1630859    .3475307
        L2D. |  -.0673661   .0469753    -1.43   0.152     -.159436    .0247039
             |
       _cons |   .0004103    .000596     0.69   0.491    -.0007578    .0015784
------------------------------------------------------------------------------
```

Results from Conventional test (Table 5.2)

```
. varnorm, skewness

   Skewness test
```

Equation	Skewness	chi2	df	Prob > chi2
D_logDTCP	-.5033	22.841	1	0.00000
D_logEX	.19629	3.474	1	0.06234
ALL		26.315	2	0.00000

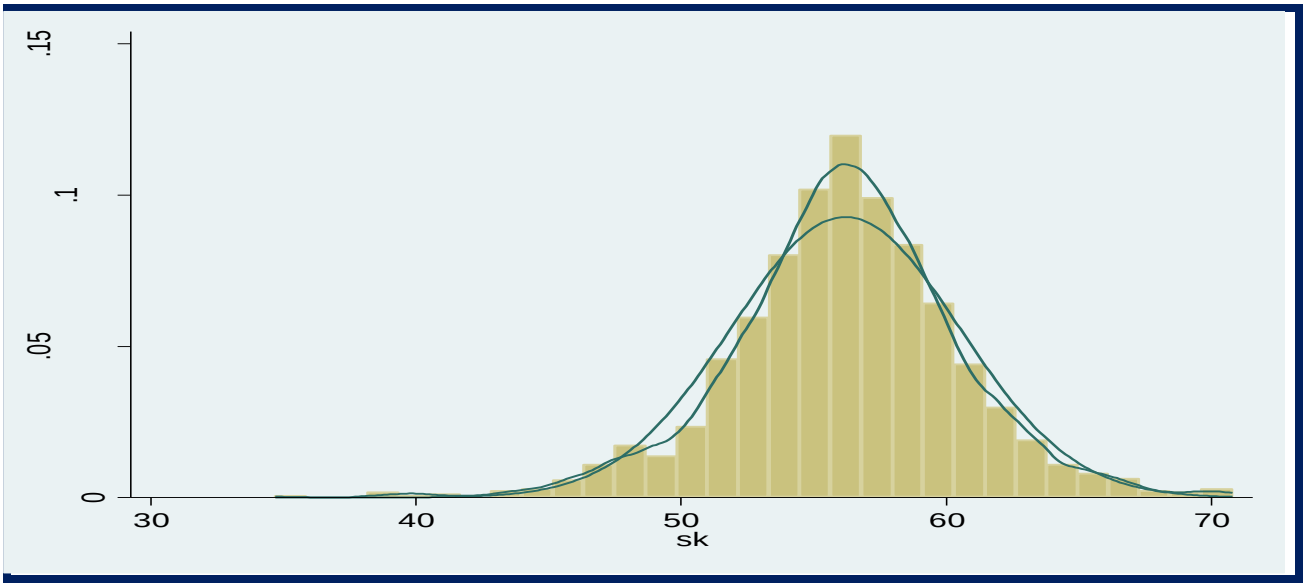
Simulation results for Skewness (Table 5.3)

```
. summarize

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
          sk |      1,500    56.19085    4.300095   34.69073   70.78248
         rej |      1,500           1           0          1          1

. histogram sk,normal kdensity
(bin=31, start=34.690735, width=1.1642498)
```

The kernel density graph for the simulation results (Table 5.4)



STATA CODES

Stata Codes for Normality test using simulation

```
//%%% Normality test using simulation-based inference%%%%%%///
clear all

import excel "C:\Users\bisma\OneDrive\Desktop\Winter\Econometrics\Module 3\Data_Project 2.xlsx", sheet("Data for Project 1 - Copy") cellrange(A1:I545) firstrow
gen logEX = ln( CanadaUSSpotExchangerate)
gen logDTCP =ln( TotalAllCommodities/ US_CPI_Index)


generate mon = tm(1972m1)+_n-1
format mon %tm
tsset mon, monthly


//#######conventional testing########//
varsoc d.logDTCP d.logEX, exo(mon)
var d.logDTCP d.logEX, lags(1/1)
varnorm, jbera

//#########bootstrapping for normality ###########//
predict rhs, xb
set seed 42457786
```

```
gen y=.
tempname jarcque
postfile `jarcque' jqb rej using resultsjqb, replace
quietly {
forvalues i=1/1500 {
replace y=rhs+rnormal(0,1)
var d.y d.logEX, lags(1/1)
varnorm, jbera
matrix v=r(jb)
scalar jqb=v[3,1]
scalar rej=(jqb> 450.580 )
post `jarcque' (jqb) (rej)
}
}
postclose `jarcque'

use resultsjqb, clear
summarize
histogram jqb,normal kdensity
```

```
/////%%%%%%%%%%%%%%LM Test for Autocorrelation-Simulation
%%%%%%%%%%%%%%//////

clear all


import excel "C:\Users\bisma\OneDrive\Desktop\Winter\Econometrics\Module
3\Data_Project 2.xlsx", sheet("Data for Project 1 - Copy") cellrange(A1:I545) firstrow

gen logEX = ln( CanadaUSSpotExchangerate)

gen logDTCP =ln( TotalAllCommodities/ US_CPI_Index)



generate mon = tm(1972m1)+_n-1

format mon %tm

tsset mon, monthly


//%%%%%%%%%%%%Conventional testing%%%%%%%%%%%%//

var d.logDTCP d.logEX, lags(1/1) exog(mon)

varlmar, mlag (3)


//%%%%%%%%%%%% Bootstrapping Approach %%%%%%%%%%%%//

set seed 576439


gen y=.

tempname Lagrangm

postfile `Lagrangm' LM rej using results, replace

quietly {

forvalues i=1/500 {
```

```
replace y=rnormal(0,1)
var d.y d.logEX, lags(1/1)
varlmar, mlag (3)
matrix v=r(lm)
scalar LM= v[1,2]
scalar rej=(LM> 7.1779)
post `Lagrangm' (LM) (rej)
}
}
postclose `Lagrangm'

use results, clear
summarize
histogram LM,kdensity
```

Stata Code for Cointegration Test

```
////%%%%%%%%%%%%%%Cointegration test using Simulation test%%%%%%%%%%%%%%////////
clear all

import excel "C:\Users\bisma\OneDrive\Desktop\Winter\Econometrics\Module 3\Data_Project 2.xlsx", sheet ("Data for Project 1 - Copy") cellrange(A1:I545) firstrow
gen logEX = ln( CanadaUSSpotExchangerate)
gen logDTCP =ln( TotalAllCommodities/ US_CPI_Index)
```

```
generate mon = tm(1972m1) +_n-1
format mon %tm
tsset mon, monthly

//////////////////////conventional test////////////////////
reg logDTCP logEX mon
predict e, residuals
varsoc e //lag selection- order criteria is based on AIC//
dfuller e, regress lag (2)

//////////////////////// Bootstrapping - Simulation ////////////////////////
set seed 48889900
scalar constant=_b[_cons]
gen y=.
gen u=.
gen resd=.
tempname dfulco
postfile `dfulco' df_test rej using results, replace
quietly{
forvalues i=1/987 {
replace u=rnormal(0,1)
replace y=constant+u
reg y logEX mon
predict e1, residuals
replace resd=e1
drop e1
```

```
dfuller resd, regress lag(2)
scalar df_test=r(Zt)
scalar rej=(df_test> -3.643)
post `dfulco' (df_test) (rej)
}
}
postclose `dfulco'

use results, clear
summarize
histogram df_test,kdensity
```

Stata Codes for Granger Causality Test

```
///////%%%%%%%%%%%%%Simulation Based Inference for Granger Causality Test%%%%%%%%%%///////
clear all

import excel "C:\Users\bisma\OneDrive\Desktop\Winter\Econometrics\Module 3\Data_Project 2.xlsx", sheet("Data for Project 1 - Copy") cellrange(A1:I545) firstrow
gen logEX = ln( CanadaUSSpotExchangerate)
gen logDTCP =ln( TotalAllCommodities/ US_CPI_Index)


generate mon = tm(1972m1)+_n-1
format mon %tm
```

```
tsset mon, monthly

//#######conventional testing#######//
varsoc logDTCP logEX, exo(mon)
regress logDTCP mon L(1/2).logDTCP L(1/2).logEX
test(L.logEX) (L2.logEX)

//#######bootstrapping######//
set seed 6743987
scalar constant=_b[_cons]
scalar b1=_b[L1.]
scalar b2=_b[L2.]


gen y=.
gen u=.
tempname Ftest
postfile `Ftest' F_test rej using results, replace
quietly {
forvalues i=1/1500 {
replace u=rnormal()
replace y=constant+b1+b2+u
regress y mon L(1/2).y L(1/2).logEX
test (L.logEX) (L2.logEX)
scalar F_test=r(F)
scalar rej=(F_test>3.18)
```

```
post `Ftest' (F_test) (rej)
}
}
postclose `Ftest'

use results, clear
summarize
histogram F_test, kdensity


//// %%%%%%%%%%%%Granger Causality - Simulation for first difference Variable %%%%%%%%//////
clear all

import excel "C:\Users\bisma\OneDrive\Desktop\Winter\Econometrics\Module 3\Data_Project 2.xlsx", sheet("Data for Project 1 - Copy") cellrange(A1:I545) firstrow
gen logEX = ln( CanadaUSSpotExchangerate)
gen logDTCP =ln( TotalAllCommodities/ US_CPI_Index)


generate mon = tm(1972m1)+_n-1
format mon %tm
tsset mon, monthly
//conventional testing//
var d.logDTCP d.logEX, lags(1/1)
vargranger
```

```
//bootstrapping//
predict rhs, xb
set seed 678544

gen y=.
tempname granger
postfile `granger' grg rej using resultsgrg, replace
quietly {
forvalues i=1/500 {
replace y=rhs+rnormal(0,1)
var d.y d.logEX, lags(1/1)
vargranger
matrix v=r(gstats)
scalar grg= v[2,1]
scalar rej=(grg> 1.5934)
post `granger' (grg) (rej)
}
}
postclose `granger'
use results, clear
summarize
histogram grg, kdensity
```

Skewness test using Simulations

```
//%%%%%%%%Skewness test using simulation %%%%%%%//
clear all

import excel "C:\Users\bisma\OneDrive\Desktop\Winter\Econometrics\Module 3\Data_Project 2.xlsx", sheet("Data for Project 1 - Copy") cellrange(A1:I545) firstrow
gen logEX = ln( CanadaUSSpotExchangerate)
gen logDTCP =ln( TotalAllCommodities/ US_CPI_Index)



generate mon = tm(1972m1)+_n-1
format mon %tm
tsset mon, monthly

///######### conventional testing #######//////
var d.logDTCP d.logEX, lags(1/2)
varnorm, skewness

/////#######bootstrapping######/////
predict rhs, xb
set seed 866567435

gen y=.
tempname skew
```

```
postfile `skew' sk rej using resultssk, replace
quietly {
forvalues i=1/1500 {
replace y=rhs+rnormal(0,1)
var d.y d.logEX, lags(1/2)
varnorm,  skewness
matrix v=r(skewness)
scalar sk=v[3,2]
scalar rej=(sk>26.315)
post `skew' (sk) (rej)
}
}
postclose `skew'

use resultssk, clear
summarize
histogram sk,normal kdensity
```